作业： 离散随机变量

- Chapter2

Problem 6. The Celtics and the Lakers are set to play a playoff series of n basketball games, where n is odd. The Celtics have a probability p of winning any one game, independent of other games.

(a) Find the values of p for which $n = 5$ is better for the Celtics than $n = 3$.

(b) Generalize part (a), i.e., for any $k > 0$, find the values for p for which $n = 2k + 1$ is better for the Celtics than $n = 2k - 1$.

Problem 7. You just rented a large house and the realtor gave you 5 keys, one for each of the 5 doors of the house. Unfortunately, all keys look identical. so to open the front door, you try them at random.

(a) Find the PMF of the number of trials you will need to open the door, under the following alternative assumptions: (1) after an unsuccessful trial. you mark the corresponding key. so that you never try it again. and (2) at each trial you are equally likely to choose any key.

(b) Repeat part (a) for the case where the realtor gave you an extra duplicate key for each of the 5 doors.

Problem 10. Form of the Poisson PMF. Let X be a Poisson random variable with parameter λ. Show that the PMF $p_X(k)$ increases monotonically with k up to the point where k reaches the largest integer not exceeding λ, and after that point decreases monotonically with k.

4. 假设一个取非负整值的随机变量X的分布列$p_k = P(X = k)$ 满足$\frac{p_n}{p_{n-1}} = \frac{\lambda}{n}, \forall n \in \mathbb{N}, n \geq 1$. 请证明$X \sim \mathcal{P}(\lambda)$，即$X$服从参数为$\lambda$的泊松分布。

5. 一个粒子在数轴上从0开始移动，每次向左或者向右移动一个单位，向右的概率为p，假设每步移动之间是相互独立的。

① 设X是粒子移动n步后所在的位置，求X的分布？

② 设D是粒子移动n步后与0的距离，求D的分布？(Hint: 可利用上问答案)

③ 若p=1/2，设粒子移动了m步之后所在的位置为随机变量Y，在“粒子移动m+n步后在j的位置”的条件下，求Y的分布。

▶ 离散型随机变量：

6. 设有一些手机共享一个WiFi点(WiFi access point)，需要通过这个WiFi点传输数据包(packets)。如果只有一个传输包，则顺利传输；如果多个手机同时传输，数据在WiFi点互相干扰，产生碰撞。我们考虑一个时间离散化的简单情形，讨论如何获得最大顺利传输率。假设共有N个手机，在第n个时刻，每个手机独立地以概率p传输数据包、概率$1-p$不传输。记$X(n)$为第n个时刻传输数据包的总个数。请问p取什么值时，每个时刻顺利传输的概率最大？

▶ 连续型随机变量：

7. 设 $X \sim N(\mu, \sigma^2)$，方程$y^2 + 4y + X = 0$无实根的概率是0.5，求μ.

8. 设$T \sim \mathcal{E}(\lambda)$. 证明其“失效率”为常数且恰为其参数
$lim_{\delta \downarrow 0} \frac{P(T \in (t, t+\delta] | T > t)}{\delta} = \lambda$.

9. 设$X \sim \mathcal{E}(\lambda)$. 证明$Y = \lambda X \sim \mathcal{E}(1)$.

10. 某工厂生产了两批数量一样多的灯泡，投放到当地市场。第一批灯泡的寿命服从0到2年的均匀分布，第二批灯泡的寿命服从参数为0.1的指数分布（单位：年）。小清从这些灯泡中随机买了一个，用了1年还没坏。请问它还能再用至少3年的概率是多少？